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Bacterial stress response: understanding the molecular mechanics to identify possible therapeutic targets

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Bacterial stress response: understanding the molecular mechanics to identify possible therapeutic targets

Abstract

Introduction: Bacteria are ubiquitous and many of them are pathogenic in nature. Entry of bacteria in host and its recognition by host defense system induce stress in host cells. With time, bacteria have also developed strategies including drug resistance to escape from antibacterial therapy as well as host defense mechanism.

Areas covered: Bacterial stress initiates and promotes adaptive immune response through several integrated mechanisms. The mechanisms of bacteria to up and down regulate different pathways involved in these responses have been discussed. The genetic expression of these pathways can be manipulated by the pharmacological interventions. Present review discusses in these contexts and explores the possibilities to overcome stress induced by bacterial pathogens and to suggest new possible therapeutic targets.

Expert opinion: In our opinion, there are two important fronts to regulate the bacterial stress. One is to target caspase involved in the process of transformation and translation at gene level and protein expression. Second is the identification of bacterial genes that lead to synthesis of abnormal end products supporting bacterial survival in host environment and also to surpass the host defense mechanism. Identification of such genes and their expression products could be an effective option to encounter bacterial resistance.

Keywords: Bacterial stress, PAMPs, PRRs, eIF-2α, ATF4, Integrated stress response, Xenophagy, Therapeutic efficacy

Article highlights

- Entry and survival of bacteria in host induce stress to the host system.
- Bacteria develop strategies to surpass host defense system.
- Resistance against antibacterial has imposed serious public health threat.
- Caspase involved in transformation and translation are potential therapeutic targets.
- Manipulation of expression of drug resistance genes can also be an effective option.

1. Introduction

Exposure to bacteria is a routine for any living system. They remain in our surroundings, food materials, habitats and even inside the body. Under such scenario, homeostasis is maintained between the predisposing factors and the body defense to remain healthy and free from bacterial infection. This homeostasis avoids intrusion of bacteria by the physical barriers or if bacteria overcome first line of defense and enters the host, then its timely neutralization or elimination of bacteria through activation of effective innate immune response [1]. Variations from health to disease state prevail depending upon several factors related to bacteria and host as type of bacteria, route of entry, number of bacteria and virulence of bacteria decide success of bacterial invasion along with physical and physiological status of host. Weak, debilitated, immunosuppressed, aged host are more vulnerable to bacterial intrusion and acquiring infection [2]. The mechanism of bacterial entry and survival in host cells involves different steps depending upon the type of bacteria as bacteria vary in size and cell wall composition including presence or absence of different appendages like capsule, fimbrie, spore and flagella [2, 3]. In simple way, it includes entry of bacteria to host cells – adaptation of bacterial cell in new host system – multiplication of bacterial cells –releasing from host cells or induction of host cell death – release from host cell – infection to adjacent cells - escape cells of host immune system – survive by gaining nutrients [1, 2]. The whole process involves the competition of bacterial cell with host cells leading to stress over host cell, often culminating in their death. Pattern of bacterial pathogenesis decide the level and type of stress induced over host system as pathological changes involve damage to cell membrane, release of cytoplasm and cytoplasmic proteins. These constitute host danger associated molecular patterns (DAMPs) as indicators of cellular/tissue damage [1] and act as messenger to induce sense of urgency to host immune system.. The other host cells recognize these indicator molecules and perceive the sense of altered homeostasis to respond and adapt in altered conditions. Response to cellular damage and adaptation to condition induces stress on host cells, mainly due to depletion of nutrients, accumulation of bacterial metabolites and host cell activities including apoptosis [1]. In this review, we will discuss all these dynamics to explore the possible targets for therapeutic use against drug resistant bacterial pathogens.

2. Entry of bacteria in host system

The entry of bacteria gets recognized by the host system to adopt strategy to prevent its entry and induce immune response to neutralize it prior to onset of infection [4]. The interaction is based on type of bacteria and its interaction with pattern recognition receptors (PRRs) present on the surface of host cells. Recognition of bacteria induces release of biomolecules like cytokines and antibacterial factors [2, 3, 5]. Entry of bacteria is prevented by first line of defense and that includes phagocytes and macrophages as front-line defense executioners along with many non-professional phagocytic cells depending upon the route of entry of pathogen as for a bacteria like *Escherichia coli* gut epithelial cells play crucial role to prevent their adhesion and also to clear them [6]. These bacteria are recognized by various receptors over host cells like PAMPs, DAMPs and PRRs that includes receptors similar to Toll, RIG-I and NOD and C-type lectin receptors [3, 5]. Detection based on these receptor lead to innate immune response and mostly results in efficient phagocytosis. It is followed by the transportation of engulfed bacteria to lysosomes and ultimately its degradation [2, 6]. During the process, phagocytes release various bio-signaling molecules like cytokines that further augment more precise and effective innate immune response and ensure clearance of large number of invaded bacteria [7]. Involvement of non-phagocytic cells as first line of defense can provide a safe niche to invader bacteria and evasively protect invader from host immune response as well as antibiotic therapy. Many bacterial intra cellular pathogens like *Brucella spp., Mycobacterium spp.* and *Coxiella burnetii* are the best examples. These can overcome host system by utilizing the mechanism of host membrane trafficking as in case of *Mycobacterium spp*. *Coxiella* are known to survive in cellular vacuoles whereas *Brucella spp.* modulates death signaling pathway of host cells and survive in replicative compartment for longer period, escaping exposure to host immune system. The immune responses to provide protection in *Brucella* vaccination is mediated by a cascade of events involving generation of reactive oxygen species and free radicals, activation of antioxidant system [4, 8].

Survival and clearance of bacteria also depends upon its route of entry in host system. Many pathogenic bacteria enter through oral route and prefer to get adsorbed to intestinal epithelium or exposed to mucosal surface of gastrointestinal tract like *E. coli*, and *Salmonella spp*. The epithelial cells of gut also recognize PAMPs and DAMPs through PRRs [1, 5]. The interaction of PRRs and patterns associated to these pathogens induces oxidative stress in endoplasmic reticulum and mitochondria. This may lead to inflammatory response and autophagy [5, 9, 10]. The pathogens particularly enterohaemorrhagic and enteropathogenic *E. coli* and other intestinal pathogens like *Salmonella spp.* and *Helicobacter spp.* also act as secondary pathogens in parasitic infections and use damages caused to epithelial structure to enter and settle down in host. The damage of cellular architecture, cytoskeleton and cellular junctions signals to promote the bacterial movement and colonization in epithelial tissues of gut [1, 6, 7, 9].

3. Mechanism of bacterial survival in host system

The level and type of stress produced by bacterial invasion depends upon the type of invading bacteria and its place to survive, grow and replicate in host cells or tissue fluids [4]. During the process, invader bacteria follow different mechanism to manipulate death pathway of the host cell [2, 11]. There are three pathways commonly used by pathogenic bacteria in human to establish pathogenesis and overcome host immune system during survival. These three pathways are: pro-death pathway, pro-survival pathway and cell death pathway that depend on activities involving mitochondria, NF-ƙB and inflammasome, respectively (Figure 1).

Mitochondria-dependent pro-death pathway is controlled by mitochondria. It modulates the process of cell death in host and mechanism of survival in apoptosis [1]. The process of apoptosis involves two pathways: a) Extrinsic pathway and b) Intrinsic pathway. These depends upon type of interaction of bacteria and host cell interaction as stimulation of death receptors like TNF-R1 and Apo2/Apo3, located on trans-membrane activate extrinsic pathway in contrast to activation of BH3 protein on release of bio-signaling molecules from the mitochondria in intrinsic pathway. BH3 proteins mask the anti-apoptotic effect of Bcl-2. In extrinsic pathway death, receptors activate death machinery through caspase-3 and -7 and lead to lysis of invader pathogens. In contrast to extrinsic pathway, intrinsic pathway is more complex and depends upon series of events after initial activation of BH3. The masking of Bcl-2 promotes oligomerization of mitochondrial outer membrane proteins (*omp*) like Bak and Bax, to initiate the release of cytochrome-c. Cytochrome-c accumulates in cytoplasm and initiates the formation of apoptosome that activate procaspase-9 and other proteolytic activities ultimately inducing apoptosis [1, 2].

The bacteria surviving initial barrier get exposed to host immune system whose activation is regulated by multiple biomolecules [7]**.** It involves activation of host inflammatory signaling and cell survival pathway. The survival of host cells depends upon timely and effectively activation of inflammatory signaling pathway [2]. Bacterial stress induced by ROS production, activation of MAP kinase and induction of Jun–N-terminal kinase (JNK) cascade leads to activation of TNF-R1. TNF-R1 regulates the protection against cell death i.e. upregulated by NF-ƙB through protective genes like Gadd45b, Mn-SOD, A20, FHC etc. Suppression of these events by protective genes regulates cell death by avoiding the activation of TNF-R1 [1, 2]. This series of event is known as NF-ƙB–dependent pro-survival pathway.

The ingress of bacterial pathogens, particularly intracellular pathogens, leads to release of exotoxins, flagellin and T3SS effectors in cytoplasm or cytoplasmic membrane of cells. The release of these bacterial components leads to change in cell surface structural design and composition inducing membrane damage and subvert cell signaling [1]. Their release in cytoplasm leads to modulation of cellular physiology and ultimately promote bacterial survival resulting in infection. The expression of bacterial proteins on host cell membrane and in cytoplasm activates innate immune system [12, 13]. The cellular PRRs, specifically NLRs recognize PAMPs and DAMPs. The recognition of PAMPs and DAMPs by NLRs initiate the process of inflammasome [5]. The composed inflammasome, comprise of bacterial cell components, NLR and ASC activate caspase-1 to release extracellular specific interleukins induced pyroptosis. The NLRs and apoptosis inhibitory proteins (NAIP) recognize different bacterial ligands and accordingly activate specific NLRC4 inflammasome as NAIP2 activates inflammasome for T3SS rod components and NAIP5 activates NLRC4 inflammasome for bacterial flagellin [1]. This pathway is designated as inflammasome dependent cell death pathway.

For the survival of bacteria, the only feasible way is either prevent or delay the process of cell death whereas, host cells or system try its best to demise the invader bacteria. This involves a series of event that induces stress over host system (Figure 1).

4. Stress induced by bacteria in host system

Invasion and survival of pathogenic bacteria in host cellular system generate oxygen-free radicals that initiate lipid peroxidation [7, 14]. Lipid peroxidation plays a critical role leading to altered homeostasis in a wide range of diseases. Further, localized or systemic bacterial infections share variable clinical manifestations that include inflammatory symptoms [8] and triggers different cytokines like IL-8, TNF-α and IL-6. These chemical mediators further induce upregulation of oxidative stress and inflammation in the tissues leading to cellular damage and perpetuate desirable immune response to maintain homeostasis [15], particularly innate immune response and subsequently adaptive immune response. Induction of protective immune potential develops when there is an excellent milieu between glutathione mediated nonenzymic antioxidant defense system and catalase produced during immunization [4, 16,17]. This process is designated as integrated stress response (ISR).

ISR is induced in response to stress stimulus of bacterial invasion and pathogenesis. It is a central dogma, comprising of numerous cascades, to overcome physiological and pathological changes leading to stress in eukaryotic cells. Both physiological as well as pathological stresses may be intrinsic and extrinsic in origin [18-20], but it is mainly extrinsic in origin, including intra intrusion of bacteria in cell cytosol or pathological changes owing to infection like derivation of glucose, amino acids, nutrients and hypoxia due to excessive bacterial growth [1]. The oxidative modifications due to bacterial stress cater both functional as well as structural changes in different enzymes and proteins that play a vital role in cell homeostasis. Likewise, altered redox potential of various transcription factors induces modification in activities of DNA binding leading to modified or altered genetic expression [14]. Such alteration activates adaptive immune response of eukaryotic cells and irrespective of bacterial virulence attributes, this system follows stress and cellular damage caused by bacterial invader in host system to induce cell recovery [1]. The central event of adaptive response in ISR is the phosphorylation of eIF-2α in eukaryotic cells to promote cellular damage [18]. It involves two way processes: a) Initiation of phosphorylation by any one or combination of different members of the eIF-2α kinase family [20] leading to remarkable decrease in global protein synthesis; b) Stimulation of genes responsible for cellular recovery like transcription factor ATF4 [1, 21]. The fundamental event of ISR, phosphorylation of eIF2α, is catalyzed by a family of four eIF2α kinases of serine/therionine [22]. Depending upon the severity and duration of stress, activation of additional factors is performed otherwise in routine phosphorylation of eIF2α downregulates global protein synthesis and upregulates ATF4.

As the stress exits, dephosphorylation of eIF2α cells returns to normal protein synthesis by termination of ISR [20, 23].

The eIF2α kinases family is constituted by four distinct kinases assigned different regulatory functions and distinct physiological and environmental stresses [19, 22]. These four kinases are: dsRNA-dependent protein kinase (PKR), PKR-like ER kinase (PERK) [18], general control nonderepressible 2 (GCN2) and heme-regulated eIF2α kinase (HRI) [19, 22]. These eIF2a kinases have one common catalytic domains and one specific regulatory domain distinct from other kinases [20]. Invading bacteria alter homeostasis either through amino acid deprivation, lysis of cell leading to heme deprivation, release of reactive oxygen species (ROS) leading to altered redox at site of infection. The stress produced by these changes initiate ISR by general control depressible 2 (GCN2), HRI and PRK like ER kinase (PERK) [19, 22,24]. Formation of dimer and autophosphorylation mediated by these kinases initiates full activation of ISR [25]. PERK activation depends upon accumulation of mis/un folded proteins in lumen of ER or direct binding of these proteins to luminal domain of ER [26]. Under these conditions, lumen bound glucose–regulated protein (GRP) is dissociated from PERK, which triggers autophosphorylation and activation of PERK [26-28]. Amino acid deprivation due to bacterial invasion initiate histidyl –tRNA synthetase domain mediated binding of GCN2 to deacylated transfer RNA (tRNA) [20, 29].

HRI is functionally distinct kinase of eIF2α family, confined to erythroid cells. It helps erythrocyte differentiation during erythropoeisis [30, 31]. Similar to other two kinases, HRI is activated by dimerization and autophosphorylation and is regulated by heme. Under routine conditions, HRI remains in inactivated dimer conformation with one heme binding domain at N-terminus and kinase insertion domain. Binding of heme at N-terminus is irreversible whereas, binding at kinase domain depends upon activated and nonactivated stage of HRI [31, 32]. HRI can be activated even without involvement of heme. The stress factors produced during bacterial pathogenesis like production of nitric oxide and ROS can induce ISR [22, 29, 31].

Other than ISR, two highly conserved signaling modules regulate stress as well as homeostasis [20]. These involve metabolic circuits essentially required for cell. These are mTOR pathway and Xenophagy [1, 9]. Xenophagy is nothing but autophagy targeting intracellular bacteria. Autophagy is essential for degradation and recycling of undesired cellular material and organelles [9] and is inhibited by the metabolic checkpoint kinase mTOR [33, 34]. Xenophagy involves host proteins that specifically target intra cellular bacteria in cells. These proteins include CALCOCO2/ NDP52, OPTN/ optineurin, SQSTM1/p62, TECPR1, SPT /septin along with ubiquitin. Xenophagy is supposed to initiate amino acid (AA) starvation caused by both extra- and intra- cellular bacteria. A study conducted on autophagy [34] has proven that the initiation of process with AA starvation leads to disintegration of mTOR from cellular membrane. Dissociation of mTOR downregulates mTOR activity and activates EIF2AK4/GCN2- EIF2S1/eIF2α/ATF3 signaling series [10, 33, 34]. The involvement of cellular proteins and signaling cascade may vary with bacteria. The entry of bacteria directly in cytosol can directly affect cellular phosphorylation process and lead to dissociation of mTOR from late endosome or lysosomes (LE/Ly) and simultaneously stimulate stress-signaling cascade EIF2AK4/EIF2S1/ATF3 [33]. The bacteria like Salmonella and Brucella, that remain confined in intra cellular compartment or vacuole like structure produce biphasic response after downregulation of mTOR and AA starvation. Salmonella based study revealed initial phase of mTOR down regulation, AA starvation and activation of ATF3 signaling pathway followed by accumulation of mTORC1 on the surface of salmonella containing vacuole suggesting normalizing conditions by replenishing amino acids from extra cellular compartments [1, 33]. All the bacterial cell constituents responsible to induce oxidative stress directly or indirectly activate immune system based on type of defense strategy adopted by host system. Further use of immunosuppressive drugs or disease conditions causing deprived immune system can also support the initiation of disease [14].

5. Potential therapeutic targets to reduce bacterial stress

Bacterial stress induces a chain of reaction with central dogma of phosphorylation of eIF2α, mediated through different kinases depending upon type of bacteria involved. The whole process of stress is initiated by stress signaling molecules or activities like deprivation of heme and amino acid, production of ROS, nitric oxide and molecules altering redox potential [20]. It is regulated by stress regulator, that control the type and duration of stress response by regulation of phosphorylation of eIF2α [22, 28, 32, 35], and finally the effector molecules like ATF-4, that lead to execution of stress response [23, 31]. These signaling molecules, regulator and effector molecules make a sequence of reaction and inhibition of any one of this string can down regulate the event (Figure 1). Further, execution of the outcome leading to genetic expression can also be inhibited. These all provide a possible pharmacological intervention based target to reduce bacterial stress [35]. The bacterial stress responses are mainly adaptive and pro-survival in nature to resolve the stress and maintain homeostasis [36]. However, prolonged bacterial stress response through activation of effector molecules leads to cell death or apoptosis or autophagy (xenophagy) [9]. There has been development of several drugs (Table 1) that target different components of cell survival and cell death pathway after bacterial stress [20, 35].

Many therapeutic drugs have been tested in cell lines for cancer therapy. Some of the drugs produce adrenergic side effects so used with precaution and under the supervision of physician. For the selection of an effective therapeutic, better understanding of different cellular process associated with stress response is required. Post eIF2α phosphorylation or formation of ATF4 is a critical point to be considered for therapeutic purpose [57]. ATF4 is a key component and is reported to regulate more than 400 genes related to cell survival mechanism and cellular death [57, 58]. The post translational modifications in ATF4 regulate gene expression so its effect of therapeutic is to be considered [34].

Understanding of physical, physiological changes produced by bacterial pathogen in cellular environment with cellular response to bacterial stress like immunomodulation, metabolism, process of cell cycle regulation, differentiation and xenophagy adjudicate a better way for the selection of effective and successful selection of therapeutics for desired targets [20]. However, still more precise and comprehensive knowledge is required to understand many pathways involved in the bacterial stress response and its arbitration. Many of the pathways discussed are proposed one and yet to be established so before the recommendation or use of any therapeutic to alter any component of bacterial stress management system utmost care should be taken to avoid any untoward or undesired side effects.

6. Expert Opinion

Bacterial stress in host cells initiates and promotes adaptive immune response through several integrated mechanisms. The central event of the integrated system is phosphorylation of eIF-2α and that is mediated by four different eIF2α kinases of serine/therionine. Phosphorylation of eIF-2α further activate effector molecules like ATF-3/4. These mechanisms have been studied in detail to up and down regulate different pathways. The ultimate genetic expression of these pathways can be manipulated by the pharmacological interventions. ATF4, a key component is reported to regulate more than 400 genes related to cell survival mechanism and cellular death. Many of these have been explored to regulate specific gene based translation process. However, many of them are still unexplored.

At present, there are two important fronts to reduce or regulate the bacterial stress. One is the use of new methodologies to target caspase involved in the transformation and translational process at gene level and expression of protein, respectively might fulfill the search of finding the pathway to control or regulate bacterial cell growth. These caspases are basically a family of protease enzymes that regulate apoptosis/ pyroptosis/ necroptosis in programmed cell death of host cells. These caspases further regulate inflammatory response that is key player in bacterial stress response. The regulation of caspases can control all cellular responses due to bacterial stress. Thus, further studies exploring mechanistic up and down regulation of caspases, their role in activation/ inhibition of specific immunological, or proinflammatory pathway with ultimate target to regulate bacterial stress response might be of great therapeutic value.

Simultaneously, the increasing bacterial resistance against commonly used antibiotic has further deepen the grievances due to longer survival of bacteria over passing routine cellular mechanism to interact with these bacteria. Second important front may be to restrict the bacterial survival in host system as in recent time some of these bacteria like methicillin resistant *Staphylococcus aureus* (MRSA), vancomycin resistant *Staphylococcus aureus* (VRSA), vancomycin resistant *Enterococcus spp*, extended-spectrum β-lactamases (ESBLs) producing bacteria, multi-drug resistant *Pseudomonas aeruginosa* (PSA), carbapenemase-producing bacteria and multi-drug resistant *Acinetobacter* (ACB) *spp*. have attained major attraction. For acquiring drug resistance, these bacteria adopt alternative metabolic pathways or drug efflux mechanism and produces mild but sustained stress over host system. Both these mechanisms lead to escape of bacteria from therapy as well as host defense mechanism. Further studies to understand the caspase based mechanism for regulation of bacterial and host cell interaction and manipulate therapeutic drug action through different pathways can throw light on the future prospects or targets of successful management of these bacterial infections adopted by these drug resistant bacteria. The bacterial resistance is usually acquired through genetic mutation and is maintained via drug resistance genes. The expression of these genes leads to synthesis of abnormal end products that misguide or surpass the host defense mechanism. Over expression of efflux proteins in the bacterial cell membrane are the most commonly employed means of therapeutic failure. Identification of such genes and their expression products finding means to silence them or discovering their alternative targets can also be lucrative options for countering resistance menace.

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* Authos have included 4 research papers and one review article. All are related to mechanism of host bacterial interaction involving interplay of prooxidant, oxidant and antioxidant.

Table 1: Potential therapeutic agents to regulate integrated stress bacterial response and their mechanism of action

Sl No.	Function	Drugs	Ref.
A. Integrated stress response inhibitor (ISRIB)			
1.	Attenuation of ATF4	ISRIB	37
B. Activator of eIF2α phosphorylation			
2.	PERK activator	CCT020312	38
3.	HRI activators	BTdCPU, N,N'-diarylureas	39
4.	GCN2 activators	Histidinol, Halofuginone, Asparaginase, Arginine deiminase	40-42
C. Inhibitors of eIF2α kinases			
5.	Inhibitor of PERK	GSK2606414 and GSK2656157, KIRA6	43-45
6.	Inhibitor of HRI kinase	Aminopyrazolindane, Indeno [1,2-c] pyrazoles	30
7.	Inhibitor of GCN2	SP600125, Indirubin-30 -monoxime, 6d, Triazolo [4,5-d] pyrimidine	46-48
D. Inhibitors of eIF2α dephosphorylation			
8.	Inhibitor of GADD34–PP1and CReP–PP1 complexes	Salubrinal	49
9.	inactivate the binding of GADD34 to PP1c	Guanabenz or its derivative Sephin1	50-51
10.	Down regulates CReP levels to decrease PP1 association with eIF2a	Nelfinavir	52
11.	Activation of ATF4 through PKR and HRI eIF2α kinases	ONC201	53
E. ATF4 post-translational modifications			
12.	RSK2 kinase inhibitor	SL0101 and its analogue	52
13.	Inhibition of ATF4 phosphorylation	H-89	55
14.	Degradation of ATF4	RPL41 protein	56

Note: Drugs enlisted are based on published literature for their mechanism of action in different pathways of integrated stress response for survival of host cells or apoptosis or autophagy. ATF-Activating transcription factor; CReP-Constitutive repressor of eIF2a phosphorylation; eIF-Eukaryotic translation initiation factor; GADD34- Growth arrest and DNA damage-inducible protein; GCN2- General control nonderepressible 2; HRI-Heme-regulated eIF2a kinase; PERK -PKR-like ER kinase; SL0101-Kaempferol-3-O-(3″,4″-di-O-acetyl-α-L-rhamnopyranoside

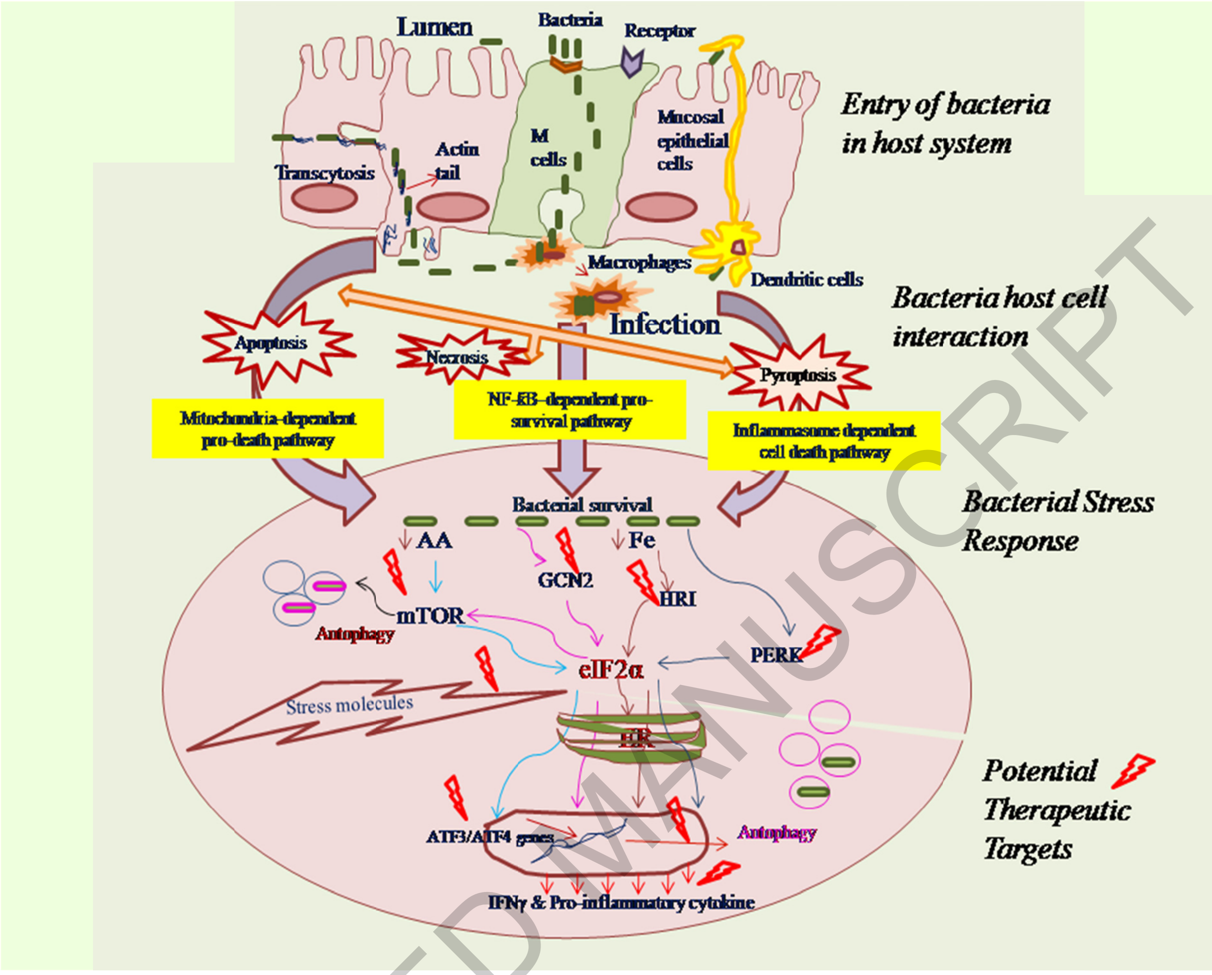

Figure 1: Schematic representation of bacterial invasion, bacteria host cell interaction, bacterial stress mechanism and potential therapeutic targets for up and down regulation of bacterial stress response. ATF- Activating transcription factor, eIF- Eukaryotic translation initiation factor; ER-Endoplasmic Reticulum; GCN2- General control nonderepressible 2; HRI - Heme-regulated eIF2α kinase; IFNγ- Interferonγ; mTOR- Mammalian target of rapamycin; NF-ĸB - Nuclear factor kappa light chain enhancer of activated B cells; PERK-PKR-like ER kinase.